SRDE-P2 | Real-Observation Portal Forecasting

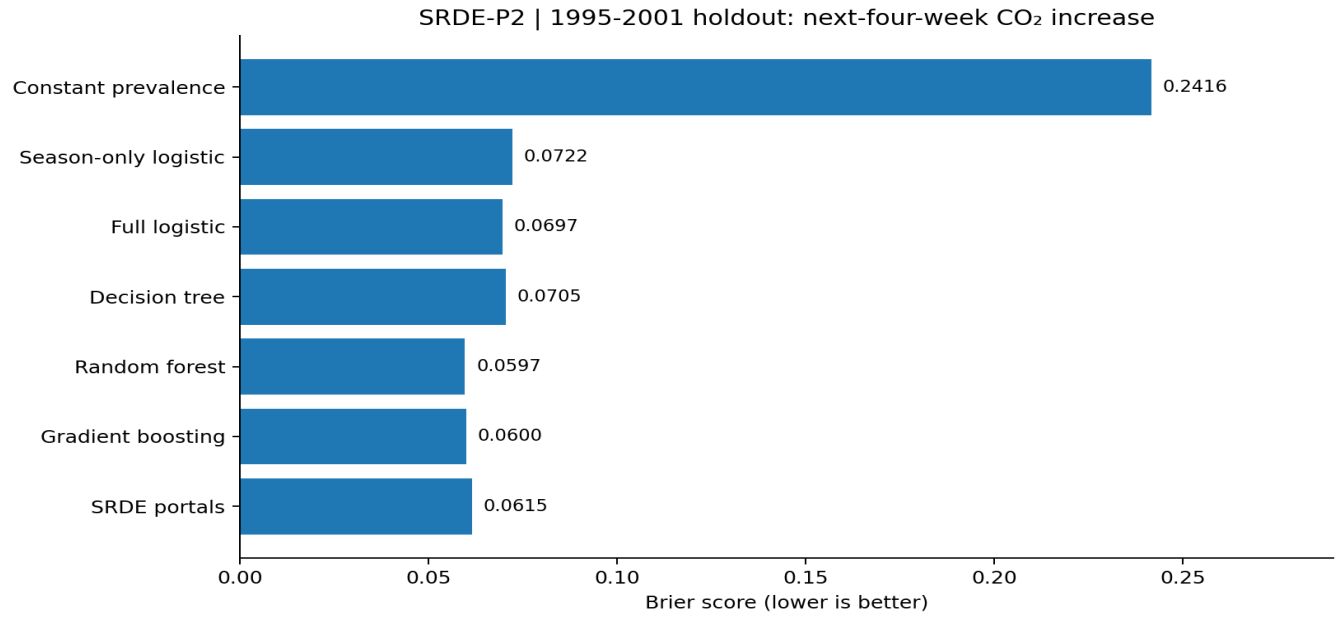
November 3, 2026 · Preregistered · Independently audited · Historical CO2 events

Outcome: The SRDE portal prototype learned useful predictive structure, but did **not** outperform the strongest conventional baselines on the held-out observations.

Measured results · 1995–2001 holdout

Method	Valid. Brier	Test Brier	Test AUC
Random forest	0.03974	0.05973	0.9737
Gradient boosting	0.03920	0.06004	0.9683
SRDE portals	0.03707	0.06150	0.9719
Full logistic	0.04655	0.06973	0.9610
Decision tree	0.04592	0.07046	0.9565
Season-only logistic	0.05531	0.07221	0.9553
Constant prevalence	0.23689	0.24159	0.5000

Primary paired difference: portal minus gradient booster = **+0.001465**; year-block 95% interval [-0.003436, +0.005951]. The interval includes zero. Random forest had the lowest test Brier at 0.059726.



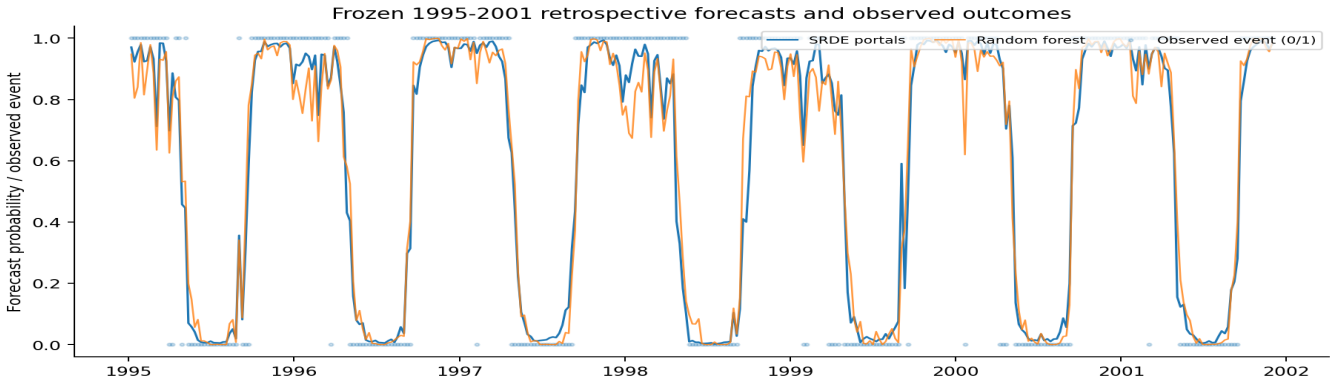
What was tested and what was held out

Historical Mauna Loa weekly CO2 (1958-2001) from the fixed statsmodels 0.14.6 dataset. Forecast: is concentration four weeks in the future greater than today? Predictors contain only historical changes and annual phase. Training: 1,320 observations ending 1988; validation: 309 ending 1994; test: 361 issued in 1995-2001. Rows whose labels crossed date boundaries were excluded.

The portal model learns recursively gated local probability models in a three-coordinate space (annual sine, cosine, recent CO2 change). **Depth cap = 100; actual depth = 3; nodes = 7.** It is similar to a model tree, not proof of exotic geometry.

ROBUSTNESS AND FAILURE MODES

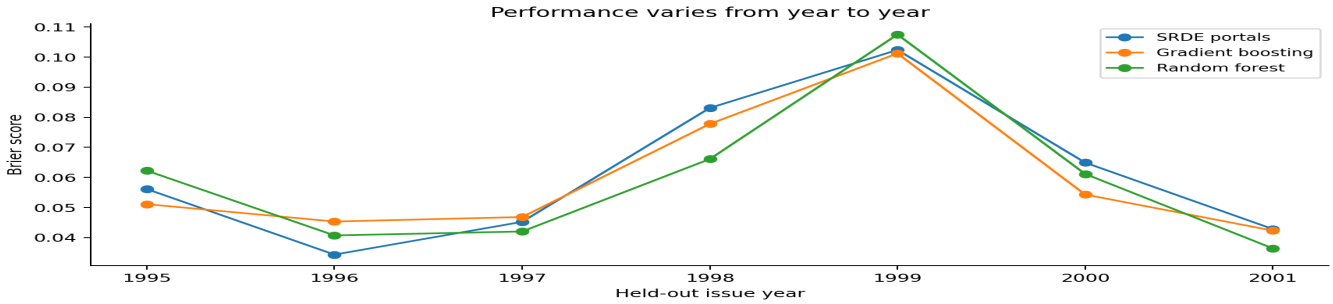
A negative result is still a result



Control checks

Shuffled-label control: Brier = 0.242079, near the constant no-information score 0.241594. **Reverse inference:** a conventional Bayesian diagnostic inferred prior trend from a known later outcome (Brier 0.161523 versus prior baseline 0.241557); this is not retrocausality.

Audit: all 361 raw events and dates were checked independently against the source; all model Brier, log-loss and accuracy scores recomputed within 1e-7; protocol and source file checksums matched the preregistered validation lock.



Scientific boundary

The model uses no emotion or feeling measurements, and provides no evidence about subconscious processes, human intelligence, universal meaning, or real-world causation. Model forecasts are retrospective, from an archival data snapshot. Only seven holdout years support the bootstrap uncertainty estimate; serial dependence remains.

Project origin: Dan Merluse. Formalization, implementation, and auditing: AI-assisted. Source scripts, full per-date forecasts, preregistration, negative controls, package versions and SHA-256 manifest are in the accompanying archive.